WORKSHEET 3.3

Slopes of Inverse Functions

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- If $g = f^{-1}$ and $f(a) = b$, then $g(b) = a$ and $g'(b) = \dfrac{1}{f'(a)}$.
- The slope of the inverse at b is the reciprocal of the slope of f at the **matching** point a, not at b.
- To find a, solve $f(a) = b$. On the exam it can always be found by inspection.
- A function with $f'(x) > 0$ everywhere is increasing, hence one-to-one, hence invertible.

Worked examples

Example 1 · From a formula

Let $f(x) = x^3 + 3x + 1$. Find $\left(f^{-1}\right)'(5)$.

Solution. $f(1) = 5$, so $f^{-1}(5) = 1$. $f'(x) = 3x^2 + 3$, so $f'(1) = 6$.

$\left(f^{-1}\right)'(5) = \dfrac{1}{6}$.

Example 2 · From values

A differentiable function f has an inverse, with $f(2) = 6$ and $f'(2) = 4$. Find $\left(f^{-1}\right)'(6)$.

Solution. The matching point is $x = 2$.

$\dfrac{1}{4}$.

Example 3 · An exponential

Let $f(x) = e^x + 2x$. Find $\left(f^{-1}\right)'(1)$.

Solution. $f(0) = 1$ and $f'(x) = e^x + 2$, so $f'(0) = 3$.

$\dfrac{1}{3}$.

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Practice

CORE

1. Let $g = f^{-1}$, with $f(4) = 9$ and $f'(4) = \dfrac{1}{2}$. Find $g'(9)$.

2. Let $f(x) = 2x - 7$. Find $\left(f^{-1}\right)'(x)$.

3. Let $f(x) = x^3$. Find $\left(f^{-1}\right)'(8)$.

4. The graph of f has slope 5 at the point $(2, 3)$. What is the slope of the graph of f^{-1}, and at which point?

EXAM LEVEL

5. Let $f(x) = x^5 + x^3 + 1$. Find $\left(f^{-1}\right)'(3)$.

6. Let $f(x) = x^3 + \ln x$ for $x > 0$. Find $\left(f^{-1}\right)'(1)$.

7. Let $f(x) = x + \sin x$. Explain what happens to $\left(f^{-1}\right)'$ at $x = \pi$.

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8. Selected values of an increasing differentiable function f are shown.

x	1	2	3
$f(x)$	4	7	12

x	1	2	3
$f'(x)$	2	5	6

Write an equation of the line tangent to the graph of f^{-1} at $x = 7$.

Question 9 · Free-response practice

Let $f(x) = x^3 + x - 4$.

(a) Explain why f has an inverse function.

(b) Find $(f^{-1})'(6)$.

(c) Write an equation of the line tangent to the graph of f^{-1} at $x = -2$.

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Worked solutions

1. $\dfrac{1}{1/2} = 2$.

2. $f'(x) = 2$ everywhere, so $\left(f^{-1}\right)'(x) = \dfrac{1}{2}$.

3. $f(2) = 8$ and $f'(2) = 12$: $\dfrac{1}{12}$.

4. Slope $\dfrac{1}{5}$, at the point $(3, 2)$.

5. $f(1) = 3$ and $f'(1) = 5 + 3 = 8$: $\dfrac{1}{8}$.

6. $f(1) = 1$ and $f'(1) = 3 + 1 = 4$: $\dfrac{1}{4}$.

7. $f(\pi) = \pi$ and $f'(\pi) = 1 + \cos\pi = 0$. The reciprocal is undefined: the graph of f^{-1} has a vertical tangent at $x = \pi$.

8. $f(2) = 7$, so the point is $(7, 2)$ and the slope is $\dfrac{1}{f'(2)} = \dfrac{1}{5}$: $y - 2 = \dfrac{1}{5}(x - 7)$.

9. (a) $f'(x) = 3x^2 + 1 > 0$ for all x, so f is increasing and therefore one-to-one.

   (b) $f(2) = 6$ and $f'(2) = 13$: $\dfrac{1}{13}$.

   (c) $f(1) = -2$ and $f'(1) = 4$. The point is $(-2, 1)$ and the slope is $\dfrac{1}{4}$: $y - 1 = \dfrac{1}{4}(x + 2)$.

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